THE THREE-COLOUR RADO NUMBERS OF $ax + 3y = 3z$

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ABSTRACT. For every integer $a \geq 4$ with $3 \nmid a$ we prove that $R_3(ax + 3y = 3z) = a^3 + a^2 + 7a + 1$, the case $b = 3$ of a conjecture of Ahmed, Zaman and Bright. The lower bound is their Theorem 1.3; this document proves the upper bound. The proof combines elementary arguments with 124 propositional unsatisfiability certificates in LRAT format, each checked by two independent proof checkers, one of them formally verified. A fixed finite system of constraints whose vertices are integer polynomials in $k = \lceil a/3 \rceil$ handles all large a simultaneously.

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1. Statement

For a linear equation $\mathcal{E}$ in three variables, the 3-colour Rado number $R_3(\mathcal{E})$ is the least positive integer N such that every colouring of $[1,N]=\{1,\dots,N\}$ with three colours contains positive integers $x,y,z\in[1,N]$ of the same colour satisfying $\mathcal{E}$. The integers x,y,z need not be distinct.

Theorem 1.1 (Ahmed–Zaman–Bright [1, Thm. 1.3])**.** *For coprime integers $a>b\geq 3$ with $a^2+a+b>b^2+ba$, $R_3(ax+by=bz)\geq a^3+a^2+(2b+1)a+1$.*

For $b=3$ the hypotheses mean $a\geq 4$ and $3\nmid a$. Ahmed, Zaman and Bright conjectured equality [1, Conj. 1.1].

Theorem 1.2. *For every integer $a\geq 4$ with $3\nmid a$,*

$$R_3(ax+3y=3z)=a^3+a^2+7a+1.$$

By Theorem 1.1 it remains to prove the upper bound:

Theorem 1.3 (upper bound)**.** *Let $a\geq 4$, $3\nmid a$, and $L=a^3+a^2+7a+1$. Every 3-colouring of $[1,L]$ contains a monochromatic solution of $ax+3y=3z$.*

Structure of the proof.

(1) $4\leq a\leq 26$: sixteen certificates (Section 3).
(2) $a\geq 13$, Lemma 4.1 (computer-assisted): the colours of $3,6,\dots,24$ form one of 53 patterns.
(3) $a\geq 23$, Lemma 5.3 (computer-assisted, 106 certificates independent of a) and Lemma 5.2: the pattern is P_0.
(4) $a\geq 7$, Section 6 (no computer): the pattern P_0 forces the colours of all multiples of 3 up to $3(a^2+a+1)$ and then L admits no colour.

The thresholds $13,23,7$ are all at most 26, so (1) and (2)–(4) together cover all $a\geq 4$ with $3\nmid a$.

2. Notation and the three families of constraints

Throughout, $a\geq 4$ and $3\nmid a$. Put

$$H=a^2+a+1,\qquad M=H-3=a^2+a-2,\qquad F=H+5=a^2+a+6,$$
$$L=1+a(H+6)=a^3+a^2+7a+1.$$

2.1. **Reduction of the equation.** If $ax+3y=3z$ then $3\mid ax$, and since $3\nmid a$ we get $3\mid x$. Writing $x=3w$ gives $z=y+aw$.

Let $c\colon[1,L]\to\{\text{three colours}\}$ be a colouring. We only look at two arithmetic progressions:

$$h(t)=c(3t)\quad(1\leq t\leq H),\qquad f(q)=c(1+aq)\quad(0\leq q\leq H+6).$$

Note $3H=3a^2+3a+3\leq L-1$, $1+aF=L-a\leq L-1$ and $1+a(H+6)=L$. Thus h and $f|_{[0,F]}$ only involve integers in $[1,L-1]$, and $f(H+6)=c(L)$.

2.2. **The constraints.** We use the following solutions of $ax+3y=3z$; in each case all three entries lie in $[1,L]$.

(BB): For integers $u\geq 1$, $s\geq 1$ with $3u\leq H$ and $s+au\leq H$: $(x,y,z)=(9u,\,3s,\,3(s+au))$. Indeed $a\cdot 9u+3\cdot 3s=3\cdot 3(s+au)$. Hence $h(3u),h(s),h(s+au)$ are not all equal. We write such a triple of h-indices as $\langle 3u,\,s,\,s+au\rangle$ and call it a *BB-triple*; its first entry is a multiple of 3 and its third entry is the second plus $a\cdot(\text{first}/3)$.

(AB): For integers $t\geq 1$, $q\geq 0$ with $t\leq H$ and $q+t\leq H+6$: $(x,y,z)=(3t,\,1+aq,\,1+a(q+t))$. Indeed $3at+3(1+aq)=3(1+a(q+t))$. Hence $h(t),f(q),f(q+t)$ are not all equal. In words: *two positions $q<q'$ of f with $f(q)=f(q')$ must not have this common colour equal to $h(q'-q)$.*

(ID): If $3s = 1 + aq$ with $1 \le s \le H$ and $0 \le q \le H + 6$, then $h(s) = f(q)$, since both are $c(3s)$.

A BB-triple may have repeated entries (for example $3u = s$); it then constrains only two values of h, which is legitimate because solutions need not consist of distinct integers.

A colouring is *solution-free* on a set S if no monochromatic solution has all three entries in S.

2.3. **Relaxations.** Each computational model below is a propositional formula whose variables encode colours of some of the integers above. Its clauses are of two kinds:

- *base clauses*: each variable takes exactly one colour, and some of the constraints (BB), (AB), (ID);
- *case clauses*, described explicitly with each model: a symmetry normalisation, a prescribed prefix, a first-occurrence condition, or a target condition (possibly built with auxiliary variables that are only required to imply their defining conditions).

Each model is used to rule out one specified class of colourings, namely the solution-free colourings of $[1, L-1]$ (resp. $[1, L]$) that satisfy its case conditions (after renaming colours where a normalisation is used). A colouring in that class satisfies every base clause, because each base clause expresses a genuine solution or the fact that each integer has one colour; it satisfies the case clauses by assumption, setting each auxiliary variable to true exactly when its defining condition holds. Hence it induces a satisfying assignment, and if the model is unsatisfiable the class is empty. That the classes ruled out in this way cover all cases is argued separately in each lemma.

In the polynomial models of Section 4 two distinct polynomials in k are treated as distinct variables even when they take equal values for some k. A real colouring assigns equal colours to equal integers, so it still induces an assignment satisfying all base clauses; the model merely omits the constraints "these two variables are equal".

2.4. **Words, prefixes and h^*.** Colours may be renamed freely. For a word $w_1 w_2 \cdots w_n$ over the three colours, its *normalisation* is obtained by renaming the colours $\mathsf{A}, \mathsf{B}, \mathsf{C}$ in order of first appearance. The *prefix* of a colouring is the normalisation of $h(1)h(2)\cdots h(8)$. Let

$$P_0 = \mathsf{AABAABAA}.$$

Let $\mathcal{B} = \{3, 6, 9, \ldots, 3a\}$ (the multiples of 3 in $[3, 3a]$) and define $h^*(t) = \mathsf{B}$ for $t \in \mathcal{B}$ and $h^*(t) = \mathsf{A}$ otherwise.

3. Small a

Lemma 3.1. *Theorem 1.3 holds for* $a \in \{4, 5, 7, 8, 10, 11, 13, 14, 16, 17, 19, 20, 22, 23, 25, 26\}$.

Proof (computer-assisted). Let $U(a) = \{3t : 1 \le t \le H\} \cup \{1 + aq : 0 \le q \le H + 6\} \subseteq [1, L]$ (its largest element is L). The formula $\mathtt{U}a$ has three Boolean variables per element of $U(a)$, the clauses "at least one colour" and "at most one colour" for each element, and, for every ordered triple $(x, y, z) \in U(a)^3$ with $ax + 3y = 3z$ and every colour, the clause "x, y, z do not all have this colour". In addition it fixes $c(1) = \mathsf{A}$ and $c(3) \in \{\mathsf{A}, \mathsf{B}\}$, which is no loss of generality after renaming colours. The formula is unsatisfiable for each listed a (Section 9). A colouring of $[1, L]$ without monochromatic solution, after renaming its colours so that $c(1) = \mathsf{A}$ and $c(3) \in \{\mathsf{A}, \mathsf{B}\}$, would restrict to a satisfying assignment. □

From now on $a \ge 27$ (the arguments below need only $a \ge 23$).

4. The polynomial window (Lemma W)

4.1. **Parametrisation.** For $a \geq 13$ with $3 \nmid a$ put $k = \lceil a/3 \rceil$ and $\rho = a - 3k \in \{-2, -1\}$. Then $a = 3k + \rho$ and $k \geq 5$. The integer k is determined by a; consequently a, $ak = 3k^2 + \rho k$ and $a(k+1) = 3k^2 + (3+\rho)k + \rho$ are polynomials in k with integer coefficients, and so is every position used in this section.

4.2. **The model $\mathcal{W}_\rho$.** *f-vertices.* The positions

$$q = \lambda a + 3kD + e,$$

for the triples (λ, D, e) listed in Table 1, regarded as integer polynomials in k after substituting $a = 3k + \rho$. Equal polynomials are identified.

λ	D	range of e
0	0	$0 \leq e \leq 7$
0	1	$-7 \leq e \leq 7$
0	2	$-7 \leq e \leq 0$
k	1	$-1 \leq e \leq 7$
k	2	$-7 \leq e \leq 7$
k	3	$-7 \leq e \leq 4$
$k+1$	1	$-2 \leq e \leq 7$
$k+1$	2	$-7 \leq e \leq 7$
$k+1$	3	$-7 \leq e \leq 4$

Table 1. The f-vertices of the window.

h-vertices. The constants $1, 2, \ldots, 8$; the heads $3, 6, 3k, 3k+3$; and every difference $q' - q$ of two f-vertices which, as a polynomial, takes values ≥ 1 for every integer $k \geq 5$.

Range. Every h-vertex takes values in $[1, H]$ and every f-vertex in $[0, F]$ for every integer $k \geq 5$. (For each quadratic this is decided exactly from its leading coefficient and the integer minimiser.) Hence every constraint below is an instance of (AB) or (BB) inside $[1, L-1]$.

Clauses.

- exactly one colour per vertex;
- (AB) for every pair of f-vertices q, q' with $q' - q \geq 1$ for all $k \geq 5$: $h(q'-q), f(q), f(q')$ not all equal;
- (BB) for every head $3u$ with $u \in \{1, 2, k, k+1\}$ and every h-vertex s such that $s + au$ (as a polynomial) is also an h-vertex: $h(3u), h(s), h(s+au)$ not all equal;
- the target: one clause, built with auxiliary variables that are only required to imply their defining conditions, stating that the colouring of $h(1), \ldots, h(8)$ uses three colours, or has $h(3) \neq h(6)$, or has as normalisation one of the eleven words

  AAAAAAAA, AAAAAAAB, AAAAAABA, AAAABAAA, AAAABABA, AAABAAAA,
  ABAAAAAA, ABBBABAB, ABBBABBB, ABBBBBAB, ABBBBBBB.

The colours of $h(1), \ldots, h(8)$ are not fixed in the model.

4.3. **The lemma.** Call a word $\mathsf{A}w_2w_3\cdots w_8$ over $\{\mathsf{A}, \mathsf{B}\}$ with $w_3 = w_6$ *admissible* if it is not one of the eleven words above. There are $2^6 = 64$ words $\mathsf{A}w_2\cdots w_8$ over $\{\mathsf{A}, \mathsf{B}\}$ with $w_3 = w_6$, hence 53 admissible words; they are listed in Appendix A, and P_0 is one of them.

Lemma 4.1 (Lemma W)**.** *Let $a \geq 13$, $3 \nmid a$, and let c be a solution-free 3-colouring of $[1, L-1]$. Then the prefix of c is an admissible word.*

Proof (computer-assisted). The prefix of a colouring that uses at most two colours on $h(1..8)$ and has $h(3) = h(6)$ is a word $\mathsf{A}w_2 \cdots w_8$ over $\{\mathsf{A}, \mathsf{B}\}$ with $w_3 = w_6$. So if the prefix is not admissible, the target clause holds. For $\rho = -2$ and $\rho = -1$ the formula $\mathcal{W}_\rho$ is unsatisfiable (Section 9). Since c induces a satisfying assignment of $\mathcal{W}_\rho$ whenever the target holds (Section 2.3), the target fails, and the prefix is admissible. □

5. From 53 prefixes to P_0

5.1. The chain lemma.

Lemma 5.1 (chain)**.** *Let $m \geq 1$ and let $g\colon [1, m] \to \{X, Y\}$ be a two-colouring with no monochromatic BB-triple inside $[1, m]$. Suppose $g(3) = X$, $k \geq 2$, $k(a + 3) + a \leq m$, and $g(3j) = X$ for $1 \leq j < k$. Then $g(3k) = X$.*

Proof. Suppose $g(3k) = Y$ and put $N = k(a + 3) = 3k + ak$. All indices below are at most $N + a \leq m$. The following are BB-triples:

triple	u	check
$\langle 3k,\ 3k,\ N\rangle$	k	$3k + ak = N$
$\langle 3(k-1),\ 3k + a,\ N\rangle$	$k - 1$	$3k + a + a(k - 1) = N$
$\langle 3,\ N,\ N + a\rangle$	1	$N + a = N + a$
$\langle 3k,\ 3k + a,\ N + a\rangle$	k	$3k + a + ak = N + a$

From the first triple, $g(N) = X$. In the second, $g(3(k - 1)) = X$ (as $k - 1 \geq 1$) and $g(N) = X$, so $g(3k + a) = Y$. In the third, $g(3) = g(N) = X$, so $g(N + a) = Y$. Now the fourth triple is monochromatic of colour Y, a contradiction. □

5.2. Two colours force P_0.

Lemma 5.2 (E4)**.** *Let $a \geq 23$, $J = \lceil (2a + 8)/3 \rceil$ and $m_0 = J(a + 3) + a$. Let c be a solution-free colouring of $[1, L - 1]$ such that h takes only two colours on $[1, m_0]$. Then the prefix of c is P_0.*

Proof. One checks $m_0 \leq H$ for $a \geq 23$ (Section 8), so h is defined on $[1, m_0]$, and it has no monochromatic BB-triple there. Let $X = h(3)$ and Y the other colour. Lemma 5.1 with $m = m_0$, applied for $k = 2, 3, \ldots, J$ in turn, gives $h(3k) = X$ for $1 \leq k \leq J$.

Let $d \in \{1, 2, 4, 5, 7, 8\}$. Since $3 \nmid d$ and $3 \nmid a$, there is $u \in \{1, 2\}$ with $3 \mid au + d$. Put $v = (au + d)/3$; then $1 \leq v \leq (2a + 8)/3 \leq J$. Now $\langle 3u,\ d,\ 3v\rangle$ is a BB-triple ($d + au = 3v \leq 3J \leq m_0$), and $h(3u) = h(3v) = X$ because $u, v \leq J$. Hence $h(d) = Y$. So $h(1) \cdots h(8) = YYXYYXYY$, whose normalisation is P_0. □

5.3. The third colour does not appear early.

Lemma 5.3 (computer-assisted, independent of a)**.** *Let $a \geq 23$, let c be a solution-free colouring of $[1, L - 1]$ whose prefix X is an admissible word, and let C be the colour not occurring in X (after the normalising renaming). Then $h(t) \neq \mathsf{C}$ for $9 \leq t \leq m_0$.*

Proof (computer-assisted). For each admissible X Appendix A gives an integer $K = K_X$ with $10 \leq K \leq 68$. Suppose C occurs in $h(9..m_0)$ and let t_0 be its first occurrence. Then either $t_0 \leq K - 1$, or $K \leq t_0 \leq m_0$; these two cases are exhaustive.

Early case, $9 \leq t_0 \leq K - 1$. Put $F_X = K + 79$. The formula $\mathtt{I}_X$ has variables for $h(1), \ldots, h(F_X)$ and $f(0), \ldots, f(F_X)$ (three colours each, exactly one colour per vertex), the unit clauses $h(1..8) = X$, the clause "$h(t) = \mathsf{C}$ for some t with $9 \leq t \leq K - 1$", and every (AB) constraint $h(t), f(q), f(q + t)$ not all equal with $1 \leq t$, $0 \leq q$, $q + t \leq F_X$. All positions satisfy $F_X \leq 147 \leq \min(H, F)$ for $a \geq 23$, so this is a relaxation, and $\mathtt{I}_X$ is unsatisfiable.

Late case, $K \le t_0 \le m_0$. By minimality of t_0, $h(t) \in \{\mathsf{A}, \mathsf{B}\}$ for $9 \le t \le K-1$. Consider the two rows $f(0), \dots, f(K-1)$ and $f(t_0), \dots, f(t_0+K-1)$. Since $t_0 + K - 1 \le m_0 + 67 \le F$ for $a \ge 23$ (Section 8), all these are values of f on $[0, F]$, and the rows are disjoint because $t_0 \ge K$. The formula $\mathtt{R}_X$ has one Boolean variable for each $h(t)$, $9 \le t \le K-1$ (colour A or B), three colour variables for each of the $2K$ cells, exactly one colour per cell, and the clauses

- for each row, each $d \in [1, K-1]$ and each $i \in [0, K-1-d]$: cells i and $i+d$ of the row are not both of colour $h(d)$ (with $h(d)$ given by X for $d \le 8$ and by the Boolean variable for $d \ge 9$); this is (AB) with difference d;
- for each $i \in [0, K-1]$: cell i of the first row and cell i of the second row are not both C; this is (AB) with difference t_0, because $h(t_0) = \mathsf{C}$.

The formula $\mathtt{R}_X$ does not depend on t_0 or a, and it is unsatisfiable.

In both cases the colouring would induce a satisfying assignment, a contradiction. □

Corollary 5.4. *Let $a \ge 23$ and let c be a solution-free colouring of $[1, L-1]$. Then the prefix of c is P_0.*

Proof. By Lemma 4.1 the prefix X is admissible, so $h(1..8)$ uses at most the two colours A, B of X. By Lemma 5.3, h does not take the third colour on $[9, m_0]$. Hence h uses at most two colours on $[1, m_0]$, and Lemma 5.2 shows that the prefix is P_0. □

6. The prefix P_0 leads to a contradiction

Throughout this section $a \ge 7$, c is a 3-colouring of $[1, L]$ without monochromatic solution, and the prefix of c is P_0. Name the colours so that $h(1..8) = \mathsf{AABAABAA}$, and let C be the third colour. Thus

$$h(3) = h(6) = \mathsf{B}, \qquad h(1) = h(2) = h(4) = h(5) = h(7) = h(8) = \mathsf{A}.$$

6.1. The third colour is absent below M.

Lemma 6.1 (nine columns)**.** *If $9 \le t \le M$ then $h(t) \ne \mathsf{C}$.*

Proof. Suppose $h(t) = \mathsf{C}$. Consider the two rows $f(0), \dots, f(8)$ and $f(t), \dots, f(t+8)$; since $t+8 \le M+8 = F$, these are defined. Index the nine columns by $i \in [0, 8]$, the cells of column i being $f(i)$ and $f(t+i)$.

(1) By (AB) with difference t and $h(t) = \mathsf{C}$, no column has C in both cells.
(2) Within a row, two cells at distance $\delta \in [1, 8]$ are not both of colour $h(\delta)$. For $\delta \in \{3, 6\}$ this colour is B; for $\delta \in \{1, 2, 4, 5, 7, 8\}$ it is A.
(3) Split the columns into the three classes $i \equiv 0, 1, 2 \pmod 3$. Two cells of one row in the same class are at distance 3 or 6, so each row has at most one B in each class. Two cells of one row in different classes are at a distance in $\{1, 2, 4, 5, 7, 8\}$, so they are not both A; hence all cells coloured A in one row lie in a single class.
(4) Therefore the cells coloured A in the two rows lie in at most two classes, and some class γ contains no A in either row. In each row the three cells of class γ contain at most one B and no A, hence at least two C. So the six cells of class γ contain at least four C, distributed over three columns; some column has C in both cells, contradicting (1). □

So h takes only the colours A, B on $[1, M]$, with $h(3) = \mathsf{B}$.

6.2. Two-colour uniqueness.

Theorem 6.2. *Let $a \ge 7$ and let $g\colon [1, M] \to \{\mathsf{A}, \mathsf{B}\}$ have no monochromatic BB-triple inside $[1, M]$, with $g(3) = \mathsf{B}$. Then $g = h^*$ on $[1, M]$.*

Proof. Step 1: $g = \mathsf{B}$ *on* $\{3, 6, \dots, 3(a-3)\}$. Apply Lemma 5.1 with $m = M$ for $k = 2, \dots, a-3$; the largest index needed is $(a-3)(a+3) + a = a^2 + a - 9 \le M$.

Step 2: $g(3(a-2)) = g(3(a-1)) = g(3a) = \mathsf{B}$. Put

$d = 3(a-2)$, $e = 3(a-1)$, $p = a+3$, $r = a^2-a+3$, $s = a^2+3$, $v = 2a-6$, $w = a^2+a-6$, $v_3 = a-6$.

All these lie in $[1, M]$ for $a \ge 7$. The following are BB-triples (the last column verifies $z = y + au$):

label	triple	u	check
T1	$\langle 3,\ 3,\ p\rangle$	1	$3 + a = p$
T2	$\langle d,\ p,\ r\rangle$	$a-2$	$a + 3 + a(a-2) = a^2 - a + 3$
T3	$\langle 3,\ r,\ s\rangle$	1	$a^2 - a + 3 + a = a^2 + 3$
T4	$\langle e,\ 3,\ r\rangle$	$a-1$	$3 + a(a-1) = a^2 - a + 3$
T5	$\langle e,\ p,\ s\rangle$	$a-1$	$a + 3 + a(a-1) = a^2 + 3$
T6	$\langle 3,\ v,\ d\rangle$	1	$2a - 6 + a = 3a - 6$
T7	$\langle d,\ d,\ w\rangle$	$a-2$	$3a - 6 + a(a-2) = a^2 + a - 6$
T8	$\langle e,\ v,\ w\rangle$	$a-1$	$2a - 6 + a(a-1) = a^2 + a - 6$
T9	$\langle 6,\ v_3,\ d\rangle$	2	$a - 6 + 2a = 3a - 6$
T10	$\langle 3a,\ v_3,\ w\rangle$	a	$a - 6 + a^2 = a^2 + a - 6$

(a) Suppose $g(d) = \mathsf{A}$. By T1 ($g(3) = \mathsf{B}$) $g(p) = \mathsf{A}$; by T2 $g(r) = \mathsf{B}$; by T3 $g(s) = \mathsf{A}$; by T4 ($g(3) = g(r) = \mathsf{B}$) $g(e) = \mathsf{A}$; now T5 is monochromatic A. Hence $g(d) = \mathsf{B}$.

(b) By T6 ($g(3) = g(d) = \mathsf{B}$) $g(v) = \mathsf{A}$; by T7 $g(w) = \mathsf{A}$; by T8, $g(e) = \mathsf{B}$.

(c) By T9 ($g(6) = g(d) = \mathsf{B}$, using $6 \in \{3, \dots, 3(a-3)\}$) $g(v_3) = \mathsf{A}$; by T10 ($g(v_3) = g(w) = \mathsf{A}$), $g(3a) = \mathsf{B}$.

So $g = \mathsf{B}$ on $\mathcal{B}$.

Step 3: $g = \mathsf{A}$ *on* $[1, M] \setminus \mathcal{B}$. Let $n \in [1, M] \setminus \mathcal{B}$. Since $3 \nmid a$ there is a unique $u_0 \in \{1, 2, 3\}$ with $au_0 \equiv n \pmod 3$.

Case $n \ge au_0 + 3$. For $u \equiv u_0 \pmod 3$ with $1 \le u \le a$, the numbers $au + 3v$ ($1 \le v \le a$) are exactly the numbers $\equiv n \pmod 3$ in the interval $I_u = [au+3,\ au+3a]$. For consecutive such u and $u+3$ the interval I_{u+3} begins at $a(u+3)+3 = (au+3a)+3$, i.e. three after the end of I_u, so the intervals I_u cover every number $\equiv n \pmod 3$ from $au_0 + 3$ up to $au_{\max} + 3a$, where $u_{\max} \ge a-2$ is the largest admissible u. Since $au_{\max} + 3a \ge a(a-2) + 3a = a^2 + a > M \ge n$, we can write $n = au + 3v$ with $1 \le u, v \le a$. Then $\langle 3u,\ 3v,\ n\rangle$ is a BB-triple with $3u, 3v \in \mathcal{B}$, so $g(n) = \mathsf{A}$.

Case $n < au_0 + 3$. Then $n \le au_0$. If $u_0 = 3$ then $n \equiv 0 \pmod 3$ and $n \le 3a$, i.e. $n \in \mathcal{B}$, which is excluded. If $u_0 = 1$ then $n \le a$ and $n + 2a \equiv a + 2a \equiv 0 \pmod 3$ with $2a < n + 2a \le 3a$, so $n + 2a \in \mathcal{B}$; the BB-triple $\langle 6,\ n,\ n+2a\rangle$ gives $g(n) = \mathsf{A}$. If $u_0 = 2$ then $n \le 2a$ and $n + a \equiv 3a \equiv 0 \pmod 3$ with $a < n + a \le 3a$, so $n + a \in \mathcal{B}$; the BB-triple $\langle 3,\ n,\ n+a\rangle$ gives $g(n) = \mathsf{A}$. □

Corollary 6.3. $h = h^*$ *on* $[1, M]$.

Proof. By Lemma 6.1 and the prefix, h is a two-colouring of $[1, M]$ by A, B with $h(3) = \mathsf{B}$; it has no monochromatic BB-triple. Apply Theorem 6.2. □

6.3. **The top of the range of h.**

Lemma 6.4 (tail)**.** $h(H-2) = h(H-1) = \mathsf{A}$ *and* $h(H) = \mathsf{C}$. *Consequently* $h = h^*$ *on* $[1, H-1]$.

Proof. An auxiliary value. Let $q_T = 11$ if $a \equiv 1$ and $q_T = 10$ if $a \equiv 2 \pmod 3$. Then $3 \mid 1 + aq_T$, and $s_T = (1 + aq_T)/3$ satisfies $3a < s_T \le M$ (as $a \ge 7$). Hence $h(s_T) = h^*(s_T) = \mathsf{A}$ and, by (ID), $f(q_T) = \mathsf{A}$.

$h(H-1) \ne \mathsf{B}$ *and* $h(H-2) \ne \mathsf{B}$. Note $H - 1 = a^2 + a$ and $H - 2 = a^2 + a - 1$.

- $\langle 3(a-2),\ 3a,\ H-1\rangle$ is a BB-triple ($u = a-2$: $3a + a(a-2) = a^2 + a$) whose first two entries lie in $\mathcal{B}$.

- If $a \equiv 1 \pmod 3$: $\langle 3a,\, a-1,\, H-2\rangle$ ($u = a$: $a - 1 + a^2$); here $a - 1 \equiv 0 \pmod 3$ and $3 \le a - 1 \le 3a$, so $a - 1 \in \mathcal{B}$.
- If $a \equiv 2 \pmod 3$: $\langle 3(a-1),\, 2a-1,\, H-2\rangle$ ($u = a - 1$: $2a - 1 + a(a-1) = a^2 + a - 1$); here $2a - 1 \equiv 0 \pmod 3$ and $2a - 1 \le 3a$, so $2a - 1 \in \mathcal{B}$.

$h(t) \neq \mathsf{C}$ *for* $t \in \{H-2, H-1\}$. Suppose $h(t) = \mathsf{C}$. Consider the rows $f(0), f(3), f(6)$ and $f(t), f(t+3), f(t+6)$; note $t + 6 \le H + 5 = F$. Column $j \in \{0, 3, 6\}$ consists of $f(j)$ and $f(t + j)$.

- By (AB) with difference t, no column has C in both cells.
- No cell is A. For the first row, the distances from q_T to $0, 3, 6$ are $\{11, 8, 5\}$ or $\{10, 7, 4\}$; they are not multiples of 3, so h has colour A at these distances, and since $f(q_T) = \mathsf{A}$, (AB) forbids A. For the second row the distances $t + j - q_T$ lie in $[a^2 + a - 12,\, a^2 + a - 4] \subseteq (3a, M]$ (as $a \ge 7$), where $h = h^* = \mathsf{A}$, and the same argument applies.
- Within a row the distances are 3 and 6, where $h = \mathsf{B}$; so each row has at most one B.

Hence each row has at least two C, four C in three columns, and some column is C in both cells: a contradiction. So $h(H-2) = h(H-1) = \mathsf{A}$; since $H - 2, H - 1 > 3a$ this agrees with h^*.

$h(H) \neq \mathsf{A}$. $\langle 3a + 3,\, 1,\, H\rangle$ is a BB-triple ($u = a + 1$: $1 + a(a+1) = H$) with $h(3a+3) = h^*(3a+3) = \mathsf{A}$ (as $3a < 3a + 3 \le M$) and $h(1) = \mathsf{A}$.

$h(H) \neq \mathsf{B}$. If $a \equiv 2 \pmod 3$: $\langle 3a,\, a+1,\, H\rangle$ ($u = a$: $a + 1 + a^2 = H$), and $a + 1 \in \mathcal{B}$. If $a \equiv 1 \pmod 3$: $\langle 3(a-1),\, 2a+1,\, H\rangle$ ($u = a - 1$: $2a + 1 + a(a-1) = H$), and $2a + 1 \in \mathcal{B}$. In both cases the first two entries lie in $\mathcal{B}$.

Hence $h(H) = \mathsf{C}$. □

6.4. L has no admissible colour.

Proposition 6.5. *There is no colour available for* L.

Proof. Let $q_0 \in \{1, 2\}$ with $3 \mid 1 + aq_0$ ($q_0 = 2$ if $a \equiv 1$, $q_0 = 1$ if $a \equiv 2 \pmod 3$), and let $Q = \{q_0 + 3j : 0 \le j \le a\}$.

The colours of f *on* Q. For $q = q_0 + 3j \in Q$ put $s_j = (1 + aq)/3 = s_0 + aj$. Then $1 \le s_j \le s_0 + a^2 \le (1 + 2a)/3 + a^2 \le M$ (as $a \ge 7$), so by (ID) and Corollary 6.3, $f(q) = h^*(s_j)$. As $3 \nmid a$, exactly one $j \in \{0, 1, 2\}$ makes s_j divisible by 3, and for it $s_j \le s_0 + 2a \le 3a$, so $f(q_0 + 3j) = \mathsf{B}$. For the other two $j \in \{0, 1, 2\}$, and for all $j \ge 3$ (where $s_j > 3a$), $f(q_0 + 3j) = \mathsf{A}$. In particular:

- one of $f(q_0)$, $f(q_0 + 3)$ is A, at a position $p_{\text{low}} \le 5$;
- $f(p_{\text{high}}) = \mathsf{A}$ for $p_{\text{high}} = q_0 + 3a$ (here $j = a \ge 3$), and $p_{\text{high}} \le 3a + 2 < H$.

The rule. If $f(q) = f(q') = \mathsf{A}$ with $1 \le q' - q \le H - 1$, then by (AB) $h(q' - q) \neq \mathsf{A}$, i.e. (by Lemma 6.4) $q' - q \in \mathcal{B}$. In particular $q' - q$ is a multiple of 3 at most $3a$.

(1) Every position $q \in [0, H + 6]$ *with* $f(q) = \mathsf{A}$ *satisfies* $q \equiv q_0 \pmod 3$. Otherwise $|q - p_{\text{high}}|$ is not a multiple of 3 and nonzero, so by the rule $|q - p_{\text{high}}| \ge H$. If $q > p_{\text{high}}$ this gives $q \ge H + p_{\text{high}} \ge H + 3a + 1 > H + 6$; if $q < p_{\text{high}}$ it gives $q \le p_{\text{high}} - H < 0$. Both are impossible.

(2) $f(q) \neq \mathsf{A}$ *for* $H \le q \le H + 5$. If $f(q) = \mathsf{A}$ then $q \equiv q_0 \pmod 3$ by (1), and $\delta = q - p_{\text{high}}$ is a multiple of 3 with $\delta \ge H - 3a - 2 = a^2 - 2a - 1 > 3a$ (as $a \ge 6$) and $\delta \le H + 5 - 3a - 1 < H$. So $\delta \notin \mathcal{B}$, contradicting the rule.

(3) One of $f(H)$, $f(H+3)$ *is* B. The positions 0 and 3 are $\not\equiv q_0 \pmod 3$, so by (1) $f(0), f(3) \in \{\mathsf{B}, \mathsf{C}\}$; they are not both B, by (AB) with difference 3 ($h(3) = \mathsf{B}$). By (2), $f(H), f(H+3) \in \{\mathsf{B}, \mathsf{C}\}$. By (AB) with difference H and $h(H) = \mathsf{C}$ (Lemma 6.4), $f(0), f(H)$ are not both C, and $f(3), f(H+3)$ are not both C. If $f(0) = \mathsf{C}$ then $f(H) = \mathsf{B}$; otherwise $f(3) = \mathsf{C}$ and $f(H+3) = \mathsf{B}$.

(4) $f(6) = \mathsf{C}$. By (1), $f(6) \neq \mathsf{A}$. Suppose $f(6) = \mathsf{B}$. Then $f(0) \neq \mathsf{B}$ (difference 6, $h(6) = \mathsf{B}$) and $f(3) \neq \mathsf{B}$ (difference 3), so $f(0) = f(3) = \mathsf{C}$, and as in (3) $f(H) = f(H+3) = \mathsf{B}$. But $f(H), f(H+3)$ are at difference 3 and $h(3) = \mathsf{B}$: a contradiction.

(5) The colour of L. Recall $c(L) = f(H + 6)$.

- $f(H+6) \neq \mathsf{C}$: (AB) with difference H between positions 6 and $H + 6$, with $h(H) = \mathsf{C}$ and $f(6) = \mathsf{C}$.

- $f(H+6) \neq \mathsf{B}$: by (3) one of $f(H), f(H+3)$ is B, at difference 6 or 3 from $H+6$, and $h(6) = h(3) = \mathsf{B}$.
- $f(H+6) \neq \mathsf{A}$: the difference $\delta = H + 6 - p_{\text{high}} = a^2 - 2a + 7 - q_0$ satisfies $3a < \delta \leq M$ (as $a \geq 7$), so $h(\delta) = h^*(\delta) = \mathsf{A}$, while $f(p_{\text{high}}) = \mathsf{A}$.

All three colours are excluded. □

7. Proof of the main theorem

Proof of Theorem 1.3. For $4 \leq a \leq 26$ this is Lemma 3.1. Let $a \geq 27$ and suppose c is a 3-colouring of $[1, L]$ without monochromatic solution. Its restriction to $[1, L-1]$ is solution-free, so by Corollary 5.4 its prefix is P_0. Section 6 (which needs $a \geq 7$) then shows that $c(L)$ admits no colour, a contradiction. □

Proof of Theorem 1.2. Theorem 1.1 gives $R_3(ax + 3y = 3z) \geq L$, and Theorem 1.3 gives $R_3(ax + 3y = 3z) \leq L$. □

8. Range conditions

The following inequalities were used above. Each is elementary; all of them, and every BB-triple identity of Sections 5 and 6, are also checked mechanically for every $a \leq 400$ with $3 \nmid a$ by `tools/check_hand_proofs.py`.

(1) $3H \leq L - 1$ and $1 + aF = L - a$ (Section 2).
(2) For $a \geq 23$: with $a = 3m+1$ ($m \geq 8$), $J = 2m+4$, $m_0 = 6m^2+23m+17$, $F = 9m^2+9m+8$ and $F - m_0 - 67 = 3m^2 - 14m - 76 \geq 4$; with $a = 3m+2$ ($m \geq 7$), $J = 2m+4$, $m_0 = 6m^2+25m+22$, $F = 9m^2+15m+12$ and $F - m_0 - 67 = 3m^2 - 10m - 77 \geq 0$. Both expressions increase with m. Also $m_0 \leq F - 67 < H$ and $147 \leq \min(H, F)$.
(3) For $a \geq 7$: $v_3 = a-6 \geq 1$; $a^2+3 \leq M$; $(a-3)(a+3)+a \leq M$; $3a < s_T \leq M$; $a^2+a-12 > 3a$; $3a + 3 \leq M$; $(1+2a)/3 + a^2 \leq M$; $a^2 - 2a - 1 > 3a$; $3a < a^2 - 2a + 7 - q_0 \leq M$.

9. Certificates

9.1. What is certified.

claim	directory in `certificates/`	number
Lemma 3.1, $4 \leq a \leq 26$	`small_a/`	16
Lemma 4.1	`lemma_W/`	2
Lemma 5.3, late case ($\mathtt{R}_X$)	`third_colour_late/`	53
Lemma 5.3, early case ($\mathtt{I}_X$)	`third_colour_early/`	53
total used in the proof		124
cross-check of Lemma 4.1	`lemma_W_alt/`	2

Every formula is in DIMACS format and comes with an LRAT proof of unsatisfiability produced by CaDiCaL [2] (commit `c6073042`) with options `--lrat --no-binary`. Each proof was checked with `lrat-check` from drat-trim (commit `2e3b2dc0`) and with the formally verified checker `cake_lpr` [4] (commit `a36874a8`). A proof checker only certifies that the given formula is unsatisfiable; that each formula encodes what is stated above is ensured by the following.

9.2. Faithfulness of the formulas.

- The formulas $\mathtt{U}a$, $\mathtt{I}_X$, $\mathtt{R}_X$ are generated by short programs (`small_a/ucert.py` and `two64.py` in the two `third_colour_*` directories) that implement the definitions of Sections 3 and 5 literally.

- For Lemma 4.1, each formula comes with a table (`*.coords.json`) giving, for every variable, its vertex as an integer polynomial in k. The program `lemma_W/audit/audit_free.py`, reading only the DIMACS file and this table, checks that every clause is (i) an exactly-one-colour clause, (ii) an (AB) clause whose three vertices satisfy $q' - q \equiv t$ as polynomials with $t \geq 1$, (iii) a (BB) clause whose vertices satisfy $z - y \equiv (3k + \rho)u$ as polynomials with $u \in \{1, 2, k, k + 1\}$, or (iv) one of the auxiliary clauses of the target; and that every vertex lies in its range for every integer $k \geq 5$. The program `concrete_free.py` instantiates every (AB)/(BB) clause for $5 \leq k \leq 60$ and checks on integers that it comes from a solution of $ax + 3y = 3z$ inside $[1, L - 1]$. The program `glue.py` enumerates all 3^8 colourings of $h(1..8)$ and, for each, decides whether the target clause together with the clauses defining its auxiliary variables is satisfiable. It checks that this is the case exactly for the colourings whose normalisation is *not* admissible; equivalently, the colourings that Lemma 4.1 leaves open are exactly those whose normalisation is one of the 53 admissible words used in Lemma 5.3.
- The cross-check formulas in `lemma_W_alt/` contain the same base clauses as the two formulas in `lemma_W/` and express the target differently (one blocking clause per allowed colouring of $h(1..8)$); they are also unsatisfiable.

9.3. **Replay and data availability.** All formulas, proofs, coordinate tables, generators and audit scripts are contained in the repository that accompanies this document, together with a SHA-256 manifest of every file. The script `certificates/replay.sh` checks the SHA-256 of every formula and runs both proof checkers on every certificate; see `README.md`. On 18 October 2026 all 126 certificates were replayed from a clean copy of the repository with the checker versions above, with no failure.

Appendix A. The 53 admissible prefixes

For each admissible word X the row length $K = K_X$ of Lemma 5.3 and the bound $F_X = K_X + 79$ of the early case.

X	K	F_X	X	K	F_X	X	K	F_X
AAAAAABB	68	147	AABBABAA	17	96	ABABAABB	15	94
AAAABAAB	60	139	AABBABAB	13	92	ABABBAAA	15	94
AAAABABB	25	104	AABBABBA	14	93	ABABBAAB	15	94
AAABAAAB	13	92	AABBABBB	14	93	ABABBABA	17	96
AAABAABA	51	130	AABBBBAA	17	96	ABABBABB	17	96
AAABAABB	14	93	AABBBBAB	13	92	ABBAABAA	16	95
AAABBAAA	41	120	AABBBBBA	14	93	ABBAABAB	15	94
AAABBAAB	14	93	AABBBBBB	13	92	ABBAABBA	16	95
AAABBABA	21	100	ABAAAAAB	27	106	ABBAABBB	15	94
AAABBABB	17	96	ABAAAABA	43	122	ABBABBAA	15	94
AABAABAA	10	89	ABAAAABB	22	101	ABBABBAB	20	99
AABAABAB	11	90	ABAABAAA	33	112	ABBABBBA	17	96
AABAABBA	13	92	ABAABAAB	25	104	ABBABBBB	24	103
AABAABBB	13	92	ABAABABA	15	94	ABBBABAA	17	96
AABABBAA	17	96	ABAABABB	16	95	ABBBABBA	17	96
AABABBAB	13	92	ABABAAAA	15	94	ABBBBBAA	17	96
AABABBBA	14	93	ABABAAAB	15	94	ABBBBBBA	44	123
AABABBBB	13	92	ABABAABA	15	94			

The eleven non-admissible words $\mathsf{A}w_2 \cdots w_8$ (with $w_3 = w_6$) are those listed in Section 4.

Use of AI tools

The proof strategy, the computer encodings, the checking scripts and the text of this document were developed with substantial assistance from AI systems (Anthropic Claude and OpenAI GPT), which also cross-checked one another's computations independently. All unsatisfiability claims rest on the certificates and checkers described in Section 9. This document has not been peer reviewed by a human expert.

References

[1] T. Ahmed, L. Zaman, C. Bright, *Symbolic sets for proving bounds on Rado numbers*, in: Proc. SC^2 2025, CEUR Workshop Proceedings Vol. 4116 (2025); arXiv:2505.12085v3.

[2] A. Biere, T. Faller, K. Fazekas, M. Fleury, N. Froleyks, F. Pollitt, *CaDiCaL 2.0*, in: CAV 2024, LNCS 14682, Springer (2024) 133–152.

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[4] Y. K. Tan, M. J. H. Heule, M. O. Myreen, *cake_lpr: verified propagation redundancy checking in CakeML*, in: TACAS 2021, LNCS 12652, Springer (2021) 223–241.